Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/09/2022
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	University Park	**State:**	PA
Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	05/31/2023
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2023 per U. Israelsson/JPL (Ed., 1/28/2020) NOTE: End date changed to 12/31/2021 per U. Israelsson/JPL (Ed., 1/8/19) NOTE: End date changed to 12/31/2018 per PI (Ed., 3/10/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1459238		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ACES (Atomic Clock Ensemble in Space) is a European Space Agency (ESA) project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for at least 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We will assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The initial working uncertainty budget projected an accuracy of 2.2×10^(-16). The proposed work will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of others, with a goal of reaching 10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify workable procedures to demonstrate its accuracy. The PI will also participate in a study on space optical clocks as a member of NASA's science definition team (SDT) for optical clocks and time transfer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on microwave cavities produced an improved design that will be used for Cs and Rb fountain clocks. These are being incorporated into new clocks at the British, Canadian, South Korean, Polish, and Indian National Laboratories, and at CERN. The new cavity design allows significantly reduced clock errors due to Doppler shifts. Our work has also led to a deeper understanding of the microwave lensing frequency shift that is being used as a correction of International Atomic Time, including the correction for the PHARAO clock. More broadly, our work contributes to scientific progress towards anticipated future optical frequency atomic clocks, including testing fundamental physics with state-of-the-art optical clocks in space.
Task Progress:	This past year, the launch of the ACES mission was extended to 2025. In the previous year, a pre-launch accuracy evaluation of the PHARAO clock was published, which completes the bulk of the Pennsylvania State University (Penn State) work expected before launch. The Penn State group continues contact with the PHARAO and ACES teams to address any issues that may arise, and maintains their software for PHARAO, for the remaining work to establish the PHARAO clock accuracy once in orbit. As a member of the NASA science definition team (SDT) for optical clocks and time transfer, the Penn State Principal Investigator (PI) Gibble participated in the development of the Fundamental physics with an Optical Clock Orbiting in Space (FOCOS) mission concept, and a paper on it was posted on arxiv.org, Gibble presented a talk on FOCOS and participated in the European Community Workshop on Cold Atoms in Space, which also resulted in a paper on arxiv.org. The SDT was recognized with a NASA Group Achievement Award for the FOCOS mission concept, “for successfully developing a mission concept for fundamental physics research in space using a high-performance optical clock and time transfer”.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Articles in Other Journals or Periodicals	Derevianko A, Gibble K, Hollberg L, Newbury NR, Oates C, Safronova MS, Sinclair LC, Yu N. "Fundamental physics with a state-of-the-art optical clock in space." arXiv:2112.10817. Posted December 20, 2021. https://doi.org/10.48550/arXiv.2112.10817 , Dec-2021
Awards	Derevianko A, Gibble K, Hollberg L, Newbury N, Oates C, Safronova MS, Sinclair L, Yu N. "NASA Group Achievement Award, May 2022." May-2022
Papers from Meeting Proceedings	Alonso I, et al. "Cold atoms in space: Community workshop summary and proposed road-map." Community Workshop on Cold Atoms in Space, Virtual. arXiv:2201.07789. Posted January 19, 2022. https://doi.org/10.48550/arXiv.2201.07789 , Jan-2022